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How Does GenAI "See" Climate Change: Exploring the Challenges and Opportunities of GenAI for Climate Visual Journalism

Saffron O'Neill[a,b], Veronica White[a,b], Tristan J. B. Cann[a,c], Felix M. Simon[d], Alastair Johnstone-Hack[e], Simon Puttock[a,c], Sylvia Hayes[a,b], Oliver Blewett[a,b], Chico Camargo[a,c,f], Travis Coan[a,g], Francisco Gonzalez Espinosa[a,c], Ranadheer Malla[a,g], Sarwat Qureshi[a,g] and Ned Westwood[a,c]

[a]Centre for Climate Change and Data Science (C3DS), University of Exeter, Exeter; [b]Geography, University of Exeter, Exeter; [c]Computer Science, University of Exeter, Exeter; [d]Oxford Internet Institute (OII) & Reuters Institute (RISJ), University of Oxford, Oxford; [e]Climate Visuals, Climate Outreach, Suite I, Windrush Court, Abingdon, Oxfordshire; [f]Department of English Language and Literature, Ewha Womans University, Seoul; [g]Politics, University of Exeter, Exeter

ABSTRACT

Visualising climate change is a challenge for journalists. Climate images can be clichéd, disconnected and even divergent from information communicated by other modalities (such as text); and can represent a missed opportunity to engage audiences. Generative AI could potentially provide transformative opportunities to visually illustrate climate news. However, risks may include reinforcing stereotyped, inaccurate or unethical visual content, with knock-on effects on audience trust and public informedness. Working as a transdisciplinary team of social scientists, computer scientists and journalists, we co-produced three studies designed to represent a spectrum of potential future engagement with generative AI technology. We prompted five generative AI tools (ChatGPT, Stable Diffusion, Midjourney, Microsoft Copilot, Generative AI by Getty Images) with seven typical climate news topics (climate change, flooding, heatwave, migration, Net Zero, COP26, nature) *via* three prompt styles (simple, detailed, story). These results indicate that there are substantial barriers for generative AI tools to create meaningful images for climate-related stories. We reflect on the implications of these results for both researchers and journalists; in terms of the evolving eye-witnessing role of photojournalism in an age of AI and misinformation, and in how such images may act to constrain transformative climate action.

KEYWORDS

Climate change; generative AI; visual communication; image; aesthetics; prompt engineering; discourse; photojournalism

Introduction

News media play a central role in shaping peoples' perceptions and behavioural engagement with climate change (Picard and Yeo 2021). Images drive audience engagement on climate change in the news media (EBU 2023), as well as peoples'

CONTACT Saffron O'Neill s.oneill@exeter.ac.uk

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emotional and behavioural responses (Graber 1990; Leiserowitz 2006). Despite the importance of climate visuals as a key device setting the frame in the news media, their role remains underdeveloped in scholarly research. Here, we are particularly focussed on the role of photographic images, and their role in the news media.

Communication is a process of meaning-making, with that meaning created in the relationship between the creator and viewer of content; a process becoming more complex with the rise of AI-generated content (Culver and Minocher 2021; Guzman 2018). Visual social semiotics is a theoretical framework for understanding meaning-making of images within a particular sociocultural context (Kress and Van Leeuwen 2020). Hall (1973) developed the concepts of denotation and connotation to explain meaning-making through imagery. Denotation is concerned with the image's "'literal' meaning" (as an analytical distinction, rather than implying a universal meaning exists) to include colour, shapes, light and a simple understanding of representation. Connotation is concerned with how what is represented in the image relate to culture, i.e., to structures of meaning which suggest additional and implied meanings. The ideological signs or symbolic messages carried by objects in an image, beyond the objects literally depicted come to have power in society, as the meaning created from images can close down debate by obscuring the possibility of alternative visions, claims, or realities (Maeselle and Raeikmaekers 2017). In a politicised digital information environment and within the context of climate denial and misinformation, the relationship between creators (machine and human) of an image, viewers of an image, and the social world in which that image exists and creates/reinforces, is of particular concern for scholarly attention; and this forms the conceptual basis of this work.

A key, if contested, function of the photographic image has been its indexicality, or its presumed function in acting as visual "proof" of the real, eye witnessing an event (Messaris and Abraham 2001). News images are presumed to represent an objective reality, rather than a socio-cultural construction of reality (Barthes 1977). This "eyewitnessing" function is of key important for photography (Zelizer 2007) and veracity of imagery particularly important for climate change, where photographs have historically acted as important documentation of environmental destruction (Doyle 2011). This is partly because calls for climate action are frequently couched in terms of scientific authority and accuracy regarding potential climate scenarios, against a backdrop of climate contrarianism (e.g. Cook et al. 2016). Therefore, if the provenance of one "climate change image" can be challenged, it may be seen to undermine the entire issue of climate change. This has played out in contentious ways in public and policy debates on climate change; for example, through a manipulated photograph of a polar bear on an ice floe undermining a message from over 200 scientists of the integrity of climate change science (O'Neill 2025). In a news context, the value of a photograph is dependent upon its presumed claim to realism (Seelig 2006), with professional practices, codes of ethics and norms generally organised around this principle (Broussard et al. 2019). The images produced by generative AI (henceforth, GenAI) tools can be aesthetically compelling and are increasingly similar to images produced by other means; for example, Nightingale and Farid (2022) discovered that participants found generated images of peoples' faces indistinguishable – and more trustworthy – than image of real faces. This raises complex questions around how we perceive truth, objectivity and the role of visuals in documenting reality (Rose 2024).

The types of images that people encounter can have different effects in terms of climate engagement. In a US, UK and Australian study, images of renewable energy engendered a sense of self-efficacy in participants, whereas imagery of politicians decreased a sense of self-efficacy (O'Neill et al. 2013). Images can also open up issues of climate justice, bringing to the fore experiences of those most vulnerable to the impacts of climate change, or can marginalise these perspectives and minimise the need for climate action (Schneider and Nocke 2014; O'Neill et al. 2023). For example, Morris (2022) found images of Hurricane María in Puerto Rico depicted Puerto Ricans as inevitable victims of the climate crisis, whilst people from the US would be "saved"; in either depiction, climate action is constructed as essentially pointless. More generally, climate change in the news media is typically visualised using a narrow range of visual tropes; drawing predominantly on images of politicians, followed by imagery of climate impacts such as droughts, floods and ice melt. Imagery of climate adaptation has been historically rare in news media coverage of climate change, despite being essential to manage climate risks (O'Neill 2013, 2025). Thus, images (as both a reflection and a construction of reality) are an important part of individual- and societal-level imaginations of sustainability, and how this might be achieved (Wright et al. 2020). The current climate visual discourse in the news media fundamentally limits formal and cultural climate politics, in terms of shaping how we might imagine and respond to climate change; and in reinforcing deeply unequal power dynamics that act to limit transformative climate action (Morris 2025).

Visual GenAI for Imagining Climate Change

One way to expand the visual discourse of climate change is, perhaps, by using GenAI tools. The term "visual GenAI" describes a range of models capable of generating novel imagery using learned patterns from existing human-generated visual data. The most popular of these are latent diffusion models, such as DALL-E and Stable Diffusion, which use learned relationships between text inputs to guide image generation. Visual GenAI covers a range of machine learning tools trained on large datasets of images and their captions.

There are already several examples of visual GenAI being used in innovative ways to expand the visual discourse of climate and sustainability topics, both within and beyond the context of climate change. In Australia, a team of lawyers used witness statements to generate images, which powerfully visualise the experiences of refugees in Australia's offshore detention centres – images that could not be created through traditional photojournalism due to lack of access (Maurice Blackburn 2023). There are emerging examples for climate and sustainability topics, too. In Panama, GenAI tools were used to help university students imagine what water scarcity might look like for familiar neighbourhoods (Chemier and Hotsko 2024). In the US, the Flood Vision project collects high-quality visual data from coastal communities which are then processed using GenAI tools to generate photorealistic visualisations of flooding and sea level rise (Climate Central 2024). In India, a project used visual GenAI in combination with art practices as a participatory research tool, to engender climate-positive future visioning (Off Centre Collective 2024). In the UK, a project highlighted the need for urgent adaptation of drainage infrastructure by using GenAI to create images of river pollution caused by extreme rainfall) (Utility Bidder 2024). Finally, the Climate Futures experiment repurposed GPT-2 as a co-designer of climate images with tuning from an input of 20 best-selling climate fiction novels (Querubín and Niederer 2024). GenAI visual content already proliferates in some digital spaces; of the images returned for the search term "climate change" on the Adobe Stock photo library, we found two-thirds (66%) of images were tagged as being AI generated (as of 13 January 2025).

However, challenges remain in how GenAI can be used to shape the visual climate discourse, including the risk of inaccurate, misleading or unethical visual content, and in reinforcing clichés, stereotypes and biases. Schofield (2024:688) has discussed how new media technologies such as visual GenAI are haunted by prior practices and communication modes. He reminds us that images have a material history, how they are "the products of a series of human actions, they don't materialise out of nowhere, even if they sometimes appear to". Visual GenAI tools utilise vast quantities of training data to generate new images. This foundational data impacts the composition and aesthetics of any images generated by the algorithm. As a result of the increasing quantity and capabilities of GenAI tools, generated images are increasingly difficult to distinguish from non-GenAI images. A key point, though, is that to the casual news consumer, GenAI images are increasingly becoming indistinguishable from photographs taken by a human photographer using a camera, and that these images are increasingly easy, quick and cheap to create. This has important implications for the role of imagery in the spread of climate misinformation (see Coan et al. 2021), as recently illustrated by the role of deepfakes in fuelling misinformation in aftermath of Hurricane Helene in the US (Arisoy 2025). Note the term "misinformation" here refers to information that is false, inaccurate or misleading (but see Treen, Williams, and O'Neill 2020 for a fuller discussion of definitions, and potential forms and impacts, of climate change misinformation).

Even if AI-generated images do not convey misinformation, they may reinforce clichéd, stereotyped or problematic visual representations; and the power relations that underlie these. GenAI tools are only as good as the data used to train them. Here lies the challenge: the world as represented through commonly employed training sets – i.e., web data including imagery and captions – is often inequitable and biased in terms of (dis)ability, race, gender and class (King 2022). These biases are therefore encoded in AI systems (Thomas and Thomson 2025). Even if the images generated are not misleading, they may well be clichéd or stereotyped. Stallabrass (2024) discusses how, when prompting DALL-E to produce an image in the style of British photographer Tish Murtha, it generates images which appear "*more archetypally like a Tish Murtha than any of her photographs*", i.e., they are exaggerated, highly conventional and uncannily similar to what has come before (Stallabrass 2024:np).

GenAI in the Newsroom

As the discussion above suggests, GenAI poses new opportunities as well as challenges for visual news making. A growing number of news organisations have started to integrate GenAI for a range of tasks (Beckett and Yaseen 2023; Simon 2025), including for content creation. Content creation, which includes visuals, is seen as a growth area. In a survey, 77% of the surveyed international media executives said content creation with AI was "very important" or "somewhat important" going forward (Newman and Cherubini 2025). At the same time, many news organisations remain cautious about the use of AI and Generative AI, including for visual purposes, given the risks that come along with the technology. These include, for example, implicit biases in the output of AI systems, so-called hallucinations (errors produced by the AI systems) as well as risks around data security, privacy, journalism ethics, and greater technological dependency (Matich, Thomson, and Thomas 2025; Misri, Blanchett, and Lindgren 2025; Moran and Shaikh 2022). More specifically, for visual tasks, Thomson, Thomas, and Matich (2024:1) found that photo editors saw potential in using AI for graphics and illustrations and to increase efficiencies but were also wary and resistant "*to AI-generated images being used in place of photojournalism*", citing "*algorithmic bias, the impact on human labour, copyright concerns, and the potential for mis/disinformation—at each stage of the production, editing, and presentation domains*" as reasons for caution.

Audiences, too, regard GenAI with a mix of interest and scepticism (Greussing et al. 2025). The balance of public opinion in a study across six countries indicated that there is relative acceptance of GenAI tools for back-end tasks in the newsroom (such as translation or editing spelling and grammar). However, there was widespread discomfort with synthetic content, including "creating an image if a real photo is not available" (Fletcher and Nielsen 2024). In terms of how content should be labelled as such if it has been produced using AI, the statement concerning GenAI of visuals engendered the highest agreement of any statement. Almost half the respondents (49%) agreed that "Creating an image if a real photograph is not available" should be labelled as such.

These initial findings suggest the public are wary of visual GenAI use in journalism. As a further example, the Norwegian Broadcasting Company, NRK, used a Midjourney image of electricity pylons against a blue sky in a story about electricity prices. The image was swiftly replaced by a stock image depicting a similar scene. Editor Espen Olsen Langfeldt commented that it was used in error (Stjern 2022). Both images were entirely consistent with the humdrum, everyday energy images used in climate coverage (O'Neill 2025), so it was presumably not the visual content that was problematic here but its AI generated nature. This makes the call from the Nieman Journalism Lab for "*rules and guidelines for creating or using AI-generated content*" and *"a broader review of how images are treated by the news media"* seem particularly pressing (Grut 2022).

This paper is concerned with how GenAI systems "see" climate change. It contributes to emerging scholarly debates on the eye-witnessing role of photojournalism; the content, meaning and role of climate visual discourse in society; and the way that AI is complicating both. It is also concerned with the wider cultural politics of climate change, through the role of photography's perceived indexicality and how this reflects on ideas of "truth" and misinformation and how this interacts with public engagement through the news media. We build on our findings to contribute practical reflections for practitioners and industry. The overarching research question for this paper is: *how do GenAI systems visualise climate change?* We examine this *via* four sub-questions:

a. How (if at all) does the **topic** influence the images produced, in terms of their denotative, connotative and ideological content?

b. How (if at all) does the choice of different **GenAI tools** influence the images produced, in terms of their content and their aesthetic qualities?
c. How (if at all) do different **prompt styles** influence the images produced?
d. What are the implications of a-c for newsrooms and researchers?

Method

Our focus is on how GenAI tools produce photorealistic images, as this image type is likely to have the most relevance to how news media organisations currently use imagery in climate content. This focus also opens up a broader conversation about how GenAI "sees" climate change, which can practically factor into the workflows and judgement calls by those considering using GenAI images versus human-produced photographs in the news media.

We use a qualitative methodology to provide rich, in-depth insights which are grounded in socio-political context. As Kessler et al. (2025) note, there is a research gap in terms of qualitative approaches to understand AI in communication. Building on the approach of Thomas and Thompson (2023) in this journal, we developed three studies, each designed to simulate how a journalist might use online tools to search for images to accompany a climate change news story (Supplementary information [SI1] provides full details of the methodology, including how the prompts were developed by our transdisciplinary team of academics and journalists). We generated images by prompting a selection of five widely available AI tools: ChatGPT, Stable Diffusion, Midjourney, Microsoft Copilot and Generative AI by Getty Images (Table 1). The first four consist of some of the most prominent tools in this space (Fletcher and Nielsen 2024) and require no (or limited) coding skills to generate images. Generative AI by Getty Images (henceforth GettyAI) is a paywalled tool designed for commercial use, representing an image source journalists will likely already be familiar with. In contrast to other tools, GettyAI claims that it is commercially clean, i.e., it owns the copyright of the images used to train its model, and therefore customers are protected (Goode 2023). Note that GettyAI is intended "to generate only creative outputs" (Getty Images, in House of Lords 2024:2re) rather than editorial content. However, it is still relevant to include as climate news does often feature creative or "stock" imagery, such as in generic images of thermometers or sunbursts to accompany heatwave news stories (O'Neill et al. 2023).

The five GenAI tools were prompted with seven climate news topics, *via* three different prompt styles (see Figure 1):

Table 1. GenAI Tools used in studies 1–3.

Tool	GenAI Model	Date of image collection	Price
MDJ - Midjourney	Midjourney Version 6.1	12.09.24	$10 per month
CGT- ChatGPT	DALL-E 3	12.09.24	$20 per month or $0.04 / image *via* API
GTY - Generative AI by Getty Images	NVIDIA Picasso/ Edify	12.09.24	£39 for 25 images
SDF - Stable Diffusion from StabilityAI	Stable Image Core	18.09.24	$0.03 per image *via* API
MCP - Microsoft Copilot	DALL-E 3	16.10.24	Free with institutional access

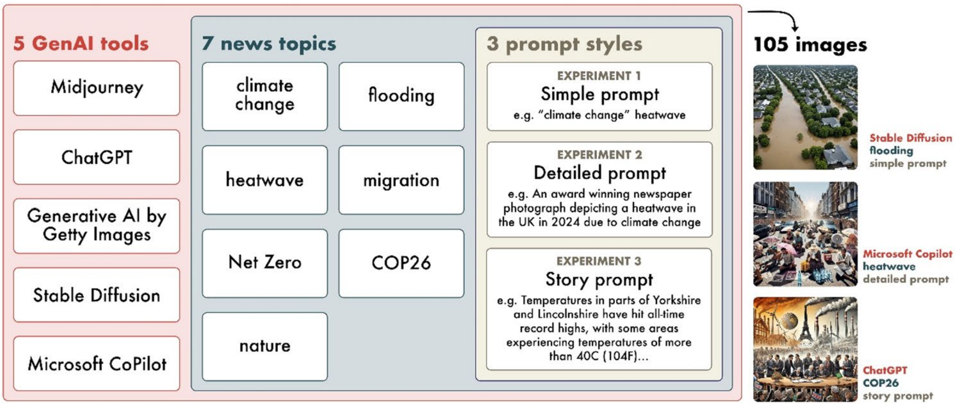

Figure 1. Schematic showing details of studies 1–3.

- The "simple" prompts (study 1) were designed to echo current journalism practice in using keyword searches via large commercial image databases. In the "simple" prompts, each topic was prompted alongside the phrase "climate change", e.g. *"climate change" flooding*.
- The "detailed" prompts (study 2) were imagined as a way to potentially enhance the keyword search, by providing detail about the style of image sought, and by adding simple geographic and time-based locators, e.g. "*an award winning newspaper photograph depicting flooding in the UK in 2024 due to climate change*" (see Supplementary information, SI2).
- The "story" prompt (study 3) used a five-paragraph extract from a representative BBC news article to provide the GenAI model with a wealth of detailed information about the topic (see Supplementary information SI3).

In total, 105 visuals were analysed as part of these studies.

Analysis

The images were analysed using visual Critical Discourse Analysis (vCDA); a reflexive approach which aims to produce rich, in-depth insights grounded in socio-cultural context. VCDA has been used to systematically analyse images to provide insights into the claim-making, framing and narratives of climate change (Hansen and Machin 2008; Rose 2023). Discourses are conceptualised as particular ways of understanding an issue, and the societal structures and power relations which are associated with these ways of engaging with the world. vCDA was carried out by the lead author, assisted in each of the topics by at least one other author who was expert in the socio-cultural context of that particular topic. As with Thomas and Thomson (2025), analysis was undertaken in an inductive fashion through an open coding process by noting salient aspects of each image: both those which emerged bottom-up from the data and top-down from coding schemas informed by previous research projects on the visual representations of climate change in the media (including O'Neill 2013; Hayes and O'Neill 2024; O'Neill et al. 2023). This top-down analysis was also informed by broader literatures on visual communication and image affordances, and on their role in meaning making, *via* an exploration of compositional elements, perspective and colour (or more precisely, *via* hue, value and saturation) (Caple 2013; Kress and van Leeuwen 2021; Rose 2023). Images were analysed individually and across topic areas by GenAI tool (Supplementary Information SI4), and study type by GenAI tool

Figure 2. An example of the images generated (from "climate change" simple, detailed and story prompts across the five genAI tools). All images generated are available in Supplementary Information (SI4 and SI5).

(Supplementary Information SI5). As an exemplar, the "climate change" prompt images can be viewed in Figure 2 below.

Results (by Topic)

Results are discussed by topic, *via* three prompt styles (simple, detailed, story). Note that due to wordcount limitations, results for climate change, flooding, migration and Net Zero topics are presented in the main paper; but heatwave, COP26 and nature topics are available in the Supplementary Information (SI6).

Climate Change

The generic "climate change" prompt (i.e., without any additional qualifiers) produces a set of 15 images which rely on clichéd and stereotyped climate visual content (see all 15 images in Figure 2). All GenAI tools include some representation of heat and burning in at least one of their outputs; such as orange-red hues, sunbursts or fire (e.g. MCP_story, CGT_simple, GTY_detailed, SDF_simple, MCP_story; Figure 2). Such representations are often interpreted, in western cultures, as suggestive of danger, heat and risk (Schneider and Nocke 2017; O'Neill et al. 2023). Similarly, all the tools except GettyAI depict either factory chimneys or billowing grey clouds (e.g. MDJ_simple). In the Stable Diffusion images, the billowing smoke assumes a more dramatic character, more akin to an explosion than to factory emissions (e.g. SDF_detailed). In news media and popular culture representations of climate change, smokestacks imagery is often untethered from recognisable geographical landmarks, which has the effect of turning them into banal and generic markers of industrial pollution, disconnected from everyday human landscapes and experience and also from systems of power, influence and lobbying which underpin the fossil fuel industry (Morris 2025; O'Neill 2025). Cracked earth, another stereotyped climate imagery signifier particularly associated with depictions of drought and extreme heat (O'Neill 2013) appears in at least one image for each GenAI tool. Perhaps one of the most iconic and recognisable of climate signifiers, the polar bear, appears in a Midjourney visual (MDJ_detailed).

The composition of image aesthetics is also intriguing in the images. For example, split-screen images offering two contrasting visions on either side of a vertical dividing line in the middle are common. The tools using DALL-E 3 (ChatGPT and Microsoft CoPilot) both produced two of their three images as split-screen visuals (see Figure 2). In all cases, one side is depicted as verdant with green trees and blue skies and the other as apocalyptic with fire, pollution and drought. The split-screen composition implies that there are different potential outcomes possible. Relatedly, two tools, Midjourney and ChatGPT, make use of globe iconography to indicate the idea of scale, choice, pathways or decision-making. Representations of the earth as a sphere hanging in space have a long history, emerging with Ptolemy's cartography but energised in environmental discourse through the NASA's Apollo Earthrise spacecraft photographs of the 1960s (Cosgrove 1994). When environmentalists claim that globe iconography is read only as epitomising the fragility of the planet and need for planetary stewardship, they ignore an alternative reading (and the inherent polysemic nature of images); as globe images can also be read as representing humans' mastery of the world (Cosgrove 1994). These two opposing readings connote very different approaches to understanding climate change, and to climate politics. Split-screen images and globe iconography also reflect common visualisations of climate change online (e.g. on Google Images, examples can be seen in Pearce and De Gaetano 2021, 5–6), highlighting the importance of training data on shaping how AI sees climate change.

It is critical to also engage with what is missing from these visual representations of climate change (Blewett et al. 2025; Rose 2024). Perhaps most obviously (and as with Pearce and De Gaetano 2021 study of Google Images) people are conspicuously absent from many of the images produced. Where humans do appear, it is in disembodied forms (e.g. a hand holding a globe, MDJ_simple; see Figure 2), as small figures lacking detail or clear facial expressions (CGT_story), or as passive, anxious faces bearing witness to a globe experiencing changes (MCP_simple). Climate communications research details how images lacking people, or "people like me", can contribute to a contested and politicised framing of climate change which disempowers people from engaging with the issue (Corner, Webster, and Teriete 2015; O'Neill 2013). Excluding humans from climate imagery, or representing them simply as passive observers, also promotes a technocentric view of climate change, which further perpetuates environmental change as an issue separate to humans and culture, and one where humans have little agency (Doyle 2011; O'Neill 2025; Pearce and De Gaetano 2021).

Taken together, the visuals may offer a literal interpretation of a warming planet, but they also represent an extreme and apocalyptic vision of what climate change might mean. GettyAI's images are noticeably different for not offering a similar apocalyptic vision. However, the GettyAI images are not linked to climate discourse in any clear way, suggesting that the GettyAI model struggles to depict an abstraction like climate change.

Flooding

The flooding images all show flood waters engulfing urban streets and homes. In several cases, the images produced are reasonably similar in content and tone to typical news media images (see particularly the MDJ_story and SDF_story images). These images show flooded streets, vehicles struggling to drive through floodwaters and sometimes rescue personnel (identifiable by their high-vis jackets and hard hats in SDF_story). In many cases there is a failure to render images realistically: from the movement of floodwater (see the wave-like water motion down each street in MDJ_detailed, or the overly ghostly and still water in MCP_story) to the place-less depiction of visual content (the models based on DALL-E 3 depict British-looking rail lines, roads and housing but the way that they are situated is not reminiscent of British urban planning). As with the climate change images, people are absent from many of these images. Where they do feature, they are either as rescue personnel or people fleeing the flood. Although some faces are turned towards the viewer (unlike the climate change images), it is still difficult to discern facial features.

A common composition in the flood images involves the use of a high-angle or birds-eye view. This has the effect of putting the viewer in an "overseer" position, viewing the scene from a detached and non-human perspective. This angle, according to Campbell (2014, 70) omits "human agency both narratively and visually", instead emphasising the overwhelming scale of destruction and flooding. Common to television coverage of flooding, aerial shots of devastated landscapes produce a disaster style spectacle in which the viewer is presented with the overwhelming scale of destruction, without any human context or experience in which to contextualise its impact. An absence of imagery depicting human adaptation and human emotions may act to privilege understanding of flooding as a remote and unavoidable issue which happens to others, elsewhere; while omitting the specific risks flooding creates in each social context and indeed how the social and political context and decision-making influences flood risk.

Despite flooding being a representational concept, GettyAI struggles to produce an image that looks much like a flooding image. Although all three GettyAI images feature water, it is only GTY_story that looks like the water might be out-of-place. GTY_simple and GTY_detailed images both feature blue skies, fluffy white clouds and lush green vegetation. Where water is pictured in these two images, it is blue, still and calm, reflecting the blue sky. It is noticeable how the GettyAI images do not use the brown hues and darker values common to so much of news imagery which depicts flooding.

Migration

Visual representations of human migration in the media often invoke several simplistic visual frames. Bleiker et al. (2013) identify the use of medium and large groups of people and a focus on boats. Johnson (2011) reflects on the growing racialisation and feminisation of the refugee and Ross and Bathia identify the "flood" frame portraying immigrants walking in a long, curved line which mimics the "bends of a river" (2021:198). These types of images have been demonstrated to shape the visual politics of migration and represent a shift away from discourses of hospitality towards a preferred policy preference for exclusion and repatriation. In the GenAI visuals, people are depicted in large groups, stretched out in long meandering lines, struggling through a hostile and barren landscape with few belongings. Few people depicted are identifiable, many with their backs turned to the viewer. Almost all people depicted are walking, but where people are using transport to move, only small boats (e.g. MJ_simple, SD_detailed) or loaded, wheeled carts (e.g. CGT_story) are in evidence. There are few references to cars, vans and lorries and an absence of infrastructure traditionally associated with migration such as trains, aeroplanes and roads.

Many of the depictions of mass migration are also racialised (notably ChatGPT and Midjourney), with few migrants appearing to be white. Depictions are also notable for how non-white migrants are represented; often poorly clothed, without shoes and carrying young children (CGT_simple CGT_story). Briggs (2003) outlines how racialised depictions of women and child refugees, in poor and vulnerable conditions produce a form of migrant imaginary which heightens traditional stereotypes to suggest vulnerability and white saviourism. This discourse is illustrated in the MCP_story image, where white people are represented in suits, coordinating a response or performing an active role. In contrast, all migrants in the background are non-white and represented as faceless, inactive and walking in various directions. Such images reproduce problematic visions of migrants as dependant, poorly resourced and unprepared people travelling from the Global South to the Global North.

Although these visuals might be consistent with many of the media narratives surrounding (mass) migration (for climate change) with a focus on international borders and an exodus of people, this is not representative of the complex evidence around climate and migration (where migration can be part of transformative adaptation to climate change and a way to build resilience; Government Office for Science 2011). By focusing on emergency mass migration, the GenAI tools function to reinforce deeply problematic and unequal power relations in terms of climate politics, rather than representing climate-induced migration as part of climate adaptation.

Net Zero

The Net Zero topic produced an incoherent set of images which seem to draw on visual clichés about sustainable futures. Wind turbines feature in all ChatGPT and Microsoft Copilot images, for example. The Earth as a globe features in SDF_simple and CGT_simple. Smokestacks feature in many images, including all Midjourney outputs. Urban greenspaces: from green walls in high-rise apartments to trees planted in busy thoroughfares, are another common visual theme. There are some cues regarding active travel (e.g. MCP_simple) but mostly, modes of transport look largely conventional with active travel (e.g. walking or cycling) and public transport largely missing from the visual outputs. The words "Net Zero" (or a jumbled approximation) featured prominently in at least one output from all the models except Midjourney.

The Net Zero topic images are evidence that a detailed visual language for the topic is yet to develop. The visual cues of wind turbines and smokestacks in the GenAI images are focussed on supply-side issues of energy transitions. This techno-optimistic representation of a Net Zero society signifies one particular type of future, and one which looks, perhaps, quite similar to current (western) society. The themes and tropes depicted in the images are reminiscent of a corporate sustainability aesthetic (Hansen and Machin 2008). The grand scale and sense of removal from everyday life represents a struggle to connect Net Zero with demand-side issues (such as charging tariffs which seek to reduce peak-time energy demand, or decreasing energy consumption for home heating through more effective household insulation). Other behaviour-change issues associated with Net Zero such as changing dietary practices or travel behaviours are also not in evidence. The images portray Net Zero as a macro scale concept involving industry, energy and technology, rather than societal or structural change towards transformative climate action (Carmichael 2019). Overall, the images largely fail to render what Net Zero might mean for people and places.

Results (by Prompt Style)

Simple Prompts

Few of the images produced in this study approximated the style or content of imagery found in current news reporting on climate change. This is because they were often artistic rather than photo-realistic in style (CGT_simple_migration), portrayed apocalyptic or fantasy-like scenes (SDF_simple_climate, MDJ_simple_NetZero) or incorporated text randomly into the image (GTY_simple_climate). Image content tended to draw on generalised visual clichés and tropes about climate and sustainability, rather than rendering a more considered view of the topic; see the recycling logo featuring prominently in a visual about Net Zero (CGT_simple_NetZero).

Detailed Prompts

The detailed prompt additions help to situate some of the image topics more concretely. In the flooding images, Midjourney automatically depicts flooding *via* the simple prompt with triple-story weatherboard houses and wide streets reminiscent of the US (MDJ_simple_flooding). When given the prompt to depict flooding "in the UK", the detailed prompt image depicts a street that looks more typically British in its architecture and urban planning (MDJ_detailed_flooding). Note that despite considerable effort spent refining the prompt process to produce an image most similar to a newspaper-style photograph (see Supplementary Information SI2), some GenAI tools still struggled with the instruction "in the style of…". Six of the eight ChatGPT images featured newspaper elements, or mocked up a newspaper page, when given this detailed prompt (rather than creating an image in the style of a newspaper image). The GettyAI model also appears to feature visual elements of a newspaper in some outputs (e.g. GTY_detailed_migration). Note that this quirk of literal prompt interpretation by the tools may not be relevant in later model iterations.

Story-Based Prompts

Images of abstract concepts do not noticeably improve when the prompt provides more detailed information, though representational concepts may do. Compare, for example, the abstract concept of "climate change" to a representational concept like "flooding". For climate change, images may become more detailed with more information (e.g. CGT_simple to CGT_story) but they are still a considerable distance from being considered a meaningful image in a news context. Similarly, although the story prompt images can become more directly related to the text provided (compare SDF_simple to SDF_story), they are not any more accurate: SDF_story depicts something of atmospheric circulation in its portrayal of an Earth from space type image (Cosgrove 1994), but does not do this in a convincing way (there are errors in rendering SDF_story, with two South Americas depicted). Conversely, when prompts become more detailed through the story prompt with representational concepts like flooding, the images do become closer to images typically used in newsrooms. For example, MDJ_simple features a flood around a US-style weatherboard house which is rendered in Midjourney's art-like house style, whereas MDJ_story features a photo-realistic style image of a recognisably British tree-lined street, red brick houses and car driving though floodwater (even including text in the corner that bears a striking resemblance to the photographer credit format used by the UK's Mail Online).

Discussion and Implications

We discuss four questions which have arisen through this paper, which are of relevance to both newsrooms and researchers:

Can GenAI Tools Help Newsrooms to Expand the Visual Discourse of Climate Change?

The introduction posited that visual GenAI might be able to assist newsrooms to make the visual discourse of climate change in the news more representative, equitable and just; for example, by picturing people and places that have been largely marginalised to date – perhaps due to difficulties of access (for example the private space of a home, for images depicting an older person experiencing heat stress during a heatwave; or imagery from a risky place for photojournalists to cover, like a conflict zone experiencing a flood event). But, GenAI is a system of probabilities, learnt mostly *via* vast datasets of content available online. It is thus unsurprising that many of the images generated in our studies reproduce problematic tropes and clichés within the bounds of current climate news coverage. And, whilst GenAI might be able to "imagine" (*via* a series of human-inputted prompts) a future scenario in a way that documentary photojournalism cannot, many of the images produced in our studies had a fairytale illustration aesthetic to them, which runs counter to guidance to feature real people and places in climate stories to support climate action (Corner, Webster, and Teriete 2015). Also, the use of GenAI images could undermine the role of correspondents and expert photojournalists who are not only are able to bring diverse visuals "from the field" to wider attention, but have the lived experience to put a visual in its narrative context.

There is the much wider issue too of model collapse: that retraining models with model-generated content causes compounding defects (Shumailov et al. 2024:755) – i.e., that GenAI requires a constant input of non-AI generated content in order to maintain output quality. Whilst some are optimistic that this means that human-created content becomes increasingly valuable in the presence of model-generated content crawled *via* the internet (e.g. Campbell 2024, 12:30), there is little evidence so far that this value will materialise in meaningful ways to, for example, visual content creators such as photojournalists.

Do GenAI Tools Offer Resource Savings to Newsrooms?

Visual GenAI offers a potentially easier, quicker and cheaper way to create custom, illustrative images for the news media sector, making it potentially appealing for resource-poor news organisations (Simon 2024). The use of such imagery potentially runs counter to editorial and AI guidelines, though, many of which impose strict rules around the generation and use of realistic AI-imagery (Becker, Simon, and Crum 2025). Although GenAI can generate images with a very low initial cost for the user, it can be very time-consuming and expertise-dependent to produce a high-quality, specified image, with different AI systems requiring different prompting strategies to achieve the desired output. Even defining a workable protocol for the studies in this paper took considerable time. Finally, audiences are sceptical of the use of AI for the creation of content, especially when being used to depict reality. Visual GenAI creates potential for misunderstanding and misinformation – with many news outlets well aware of the potential reputational risk of using AI (Newman and Cherubini 2025, 30).

What about Environmental Considerations for Newsrooms in Using Visual GenAI?

There is an increasing awareness of the environmental cost of GenAI. This environmental cost is particularly pronounced for image generation (in comparison to text generation): Luccioni, Jernite, and Strubell (2024), for example, have calculated that generating just one image can use a similar amount of energy to charging a smartphone halfway. As models become more complex and powerful, they have also become more energy intensive. There are other environmental costs to GenAI too, notably the large quantities of water used for data storage facilities (Li et al. 2023). If many images are generated and discarded in pursuit of one final image, this could lead to a substantial environmental cost.

There are some confounding factors to consider here, though. Photojournalism can also be an environmentally-costly pursuit, particularly when news organisations fly in non-local photographers to cover a story. For example, transportation, especially air travel, has a very substantial carbon footprint (Whitmarsh et al. 2020). It is certainly not straightforward to ascertain that GenAI is environmentally "bad" compared to the status quo. Rephrasing Miller (2015:660) as he pointedly commented in an essay in this journal: "I am uncertain whether a heavy reliance on carbon-fuelled [GenAI] is preferable or inferior to gumshoe and tarmac journalism […] I am, however, clear in my call for a comparative audit of the impact of these forms of journalistic research".

What Are the Wider Implications for Newsrooms, Researchers and Society?

Our findings contribute to scholarship on the evolving eye-witnessing role of photojournalism in an age of AI and misinformation, as well as critical literatures on the limitations of the dominant visuals of climate change. In terms of truth, proof and witnessing, concerns about GenAI imagery's effect goes two ways: a GenAI image may misrepresent a real-world event (see discussion of Hurricane Helene above; Arisoy 2025), but also, public awareness of the increasing ability of GenAI tools to create photorealistic images may lead to increased scepticism of *any* news images (even those not generated by AI). Either may further reduce public trust in news. We have demonstrated how GenAI images of climate change reproduce not only problematic visual representations, but also represent and reinforce existing power structures in societies. These representations discursively obscure or constrain the possibilities to imagine alternative and potentially transformative responses to climate change.

Whether and how photojournalism is threatened by digital technologies has seen considerable debate (Bock 2012; Matich, Thomson, and Thomas 2025; Moran and

Shaikh 2022; Simon 2025). It has always been possible to distort, mislead and manipulate visual content or its meaning. The ease of generating photorealistic imagery *via* GenAI makes it more important than ever to increase awareness that images are always a form of mediated communication which require scrutiny and a critical lens (Newton 1998:8). This study therefore speaks to emerging scholarly calls in journalism to take AI hype seriously (Dodds, Zamith, and Lewis 2025; Lewis, Zamith, and Bunquin 2025) and view AI as a constitutive moment for scholarly work to critique and explore the social and cultural factors of technological change.

Conclusion

Imagery is a powerful tool for climate communication, and recent advances in GenAI may create new ways of visually communicating one of humanity's greatest challenges. Yet, as our work shows, there are currently substantial barriers for GenAI tools to create meaningful images for climate-related news stories. Given the way that GenAI models are trained and the data that they are (and will continue to be) trained on, problematic imagery will likely continue to be produced. Hopes that the use of GenAI will improve visual communication of climate change in the news media have to be doubted, for now. The varying quality of the images produced, their unclear efficacy with audiences, the potential environmental costs and significant ethical challenges involved, and the scepticism (and reluctance) of both audiences and the industry when it comes to their use in news all present challenges. Future research could usefully offer insights into how both journalists and audiences use and interpret visual GenAI content about climate change; for example, through newsroom ethnographic studies and survey methodologies.

Nevertheless, it does not follow that we can therefore consider this case closed. Much will depend on the further development of GenAI tools and as of yet unforeseen uses within the news media. In addition, other sectors, who face different incentives (such as the entertainment or advertising industries) will likely make use of these tools to communicate climate change, where audiences willingly spend time and consume information (consider movies, games or time spent on social media). How and what the adoption and use of GenAI systems for climate change-related imagery will mean in these contexts will be a fruitful topic of further study.

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Author Contributions

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editing; **Alastair Johnstone-Hack**: Conceptualization, Methodology, Writing – review & editing; **Simon Puttock**: Investigation, Writing – review & editing; **Sylvia Hayes**: Conceptualization, Investigation, Writing – review & editing; **Oliver Blewett**: Investigation, Writing – review & editing; **Chico Camargo**: Writing – review & editing; **Travis Coan**: Conceptualization, Writing – review & editing; **Francisco Gonzalez Espinosa**: Writing – review & editing; **Ranadheer Malla**: Writing – review & editing; **Sarwat Qureshi**: Methodology, Writing – review & editing; **Ned Westwood**: Writing – review & editing.

Data Deposition

Data from this study is available at doi.org/10.6084/m9.figshare.28370261.

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ORCID

Felix M. Simon http://orcid.org/0000-0002-0371-4653

Data Availability Statement

The authors confirm that the data supporting the findings of this study are available within the article and its supplementary materials.

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